A COMPLETE RING WITH A PRINCIPAL IDEAL WHOSE COMPLETED LOCALIZATION IS NOT FLAT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The counted proof is the two-line instantiation; the load-bearing non-flatness lives complete in the context span (kernel comparison + integrality absurdity); 'observe' bridges via R[[x]]_f = R_f[[x]] and x-adic completeness of power series rings.

Editorial note. The label lemma-nonflat-completion-localization names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 1103-1140: with R = k[y, z, a_1, a_2, ...]/(ya_i, a_i a_j) and f the residue class of z, the ring map R[[x]] -> R_f[[x]] is not flat - the kernels I of y : R[[x]] -> R[[x]] and J of y : R_f[[x]] -> R_f[[x]] satisfy J != I R_f[[x]], proved by comparing the coefficient of a_n after reduction mod (y, a_1, ...), Noetherian finiteness in k[z][[x]], and Algebra lemma-characterize-integral-element giving the absurd integrality of z^{-1}x; flatness would force J = I R_f[[x]] by tensoring 0 -> I -> R[[x]] -> R[[x]] -> 0
- x-adic completion of power series rings: R[[x]] is x-adically complete, and localizing at a coefficient-ring element commutes with power series: R[[x]]_f = R_f[[x]]

Source reference. Original lemma 052I: sources/stacks-chow/tags.txt, source lines 6571–6571. Prerequisite pointer invoked at 1183; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Next, we will construct an example where the completion of a localization is nonflat. To do this consider the ring

$$R = k[y, z, a_1, a_2, a_3, \ldots]/(ya_i, a_i a_j)$$

Denote $f \in R$ the residue class of z. We claim the ring map

$$R[[x]] \longrightarrow R_f[[x]] \tag{0.0.1}$$

isn't flat. Let I be the kernel of $y : R[[x]] \to R[[x]]$. A typical element g of I looks like $g = \sum g_{n,m} a_m x^n$ where $g_{n,m} \in k[z]$ and for a given n only a finite number

Date: stacks-project:examples.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

of nonzero $g_{n,m}$. Let J be the kernel of $y : R_f[[x]] \to R_f[[x]]$. We claim that $J \neq IR_f[[x]]$. Namely, if this were true then we would have

$$\sum z^{-n} a_n x^n = \sum\nolimits_{i=1,\ldots,m} h_i g_i$$

for some $m \geq 1$, $g_i \in I$, and $h_i \in R_f[[x]]$. Say $h_i = \bar{h}_i \bmod (y, a_1, a_2, a_3, \ldots)$ with $\bar{h}_i \in k[z, 1/z][[x]]$. Looking at the coefficient of a_n and using the description of the elements g_i above we would get

$$z^{-n} x^n = \sum \bar{h}_i \bar{g}_{i,n}$$

for some $\bar{g}_{i,n} \in k[z][[x]]$. This would mean that all $z^{-n}x^n$ are contained in the finite $k[z][[x]]$-module generated by the elements $\bar{h}_i$. Since $k[z][[x]]$ is Noetherian this implies that the $R[z][[x]]$-submodule of $k[z, 1/z][[x]]$ generated by $1, z^{-1}x, z^{-2}x^2, \ldots$ is finite. By Algebra, Lemma 052I we would conclude that $z^{-1}x$ is integral over $k[z][[x]]$ which is absurd. On the other hand, if (0.0.1) were flat, then we would get $J = IR_f[[x]]$ by tensoring the exact sequence $0 \to I \to R[[x]] \xrightarrow{y} R[[x]]$ with $R_f[[x]]$.

Lemma 0.1. *There exists a ring A complete with respect to a principal ideal I and an element $f \in A$ such that the I-adic completion $A_f^\wedge$ of A_f is not flat over A.*

Proof. Set $A = R[[x]]$ and $I = (x)$ and observe that $R_f[[x]]$ is the completion of $R[[x]]_f$. □